

Transforming Feynman Diagrams and Correlation Functions as GKZ Systems

Matao MacGoey¹

¹Department of Physics, University of Bath, Bath BA2 7AY, UK

E-mail: mm3517@bath.ac.uk

Abstract

This report introduces Gel'fand-Kapranov-Zelevinsky (GKZ) systems and the mathematical structures associated with them, in view of applications to Feynman diagrams and correlation functions. We analyse the triangle Feynman diagram and a multi-Bessel (triple K) integral, and extend to additional examples in two- and four-dimensional parameter spaces. Using these worked examples, we explore the interplay between the integral and polytope sides of GKZ systems. Through the lens of affine transformations, we find subtle inconsistencies between these perspectives. Underpinning this work, we develop an open-source Python library capable of constructing and deriving GKZ systems arising from Feynman diagrams, providing computational tools to explore these structures systematically.

1 Introduction

Evaluating scattering amplitudes is a fundamental problem in different areas of physics, from particle physics to cosmology. These perturbative methods require computing high-order Feynman diagrams and correlation functions, which is complex due to, among other factors, high-dimensional integration, renormalisation, and divergences.

In 1963, Regge conjectured that all Feynman integrals belong to a generalisation of hypergeometric functions, and therefore obey an analogous system of differential equations [1, 2]. More recently, this connection is elucidated by parameterising Feynman integrals in Lee-Pomeransky (LP) form [3], and promoting the kinematic coefficients to indeterminate variables. At this point, the integral is now in the form of an $\mathcal{A}$ -hypergeometric function, also known as a GKZ integral. These functions are well-known in mathematics and they obey differential equations, which form a GKZ system [4]. Associated with the GKZ system is a particular Newton polytope, which is critical component part of understanding affine transformations.

The study of Feynman integrals, conformal field theory (CFT) correlators and cosmological correlators as GKZ integrals is a recent and developing area of study. In this report, we examine affine equivalences between the polytopes representing the GKZ systems of Feynman integrals and correlation functions. These are of interest due to a result that the integrals of two affinely equivalent GKZ systems are proportional [5]. In theory, this would allow us to bypass computationally-intensive calculations by relating some complicated, high-order Feynman integral or correlation function, to a simpler Feynman integral, or even to an integral with a known analytical solution.

One such example, identified by Caloro and McFadden [6] relates the triangle Feynman diagram to a multi-Bessel function integral, known as the triple K, which is a correlation function that arises from CFT and cosmology. In this report, we generalise the results from the triangle-triple K relation to understand how GKZ systems can be equivalent in a general sense. We develop a new open-source Python library designed for computations related to GKZ and Feynman integrals. As a result of our investigation, we find evidence to suggest that there is a slight disconnect between the integral and polytope side of GKZ.

2 GKZ hypergeometric systems

In this section, we will cover the basics of GKZ systems and how to extract their related data. Rather than deriving the properties of GKZ, we will show why they exist and how to interpret them. For a comprehensive overview of GKZ theory, see [3, 4, 7–9].

2.1 Building the GKZ structure

Within the context of this report, the most general form of a GKZ integral is of the form

$$I = \int_0^\infty dz \frac{\prod_{i=1}^n z_i^{\gamma_i - 1}}{\mathcal{D}(\mathbf{x}, \mathbf{z})^{\gamma_0}}, \quad (2.1)$$

where $\gamma = (\gamma_0, \gamma_1, \dots, \gamma_n)$ are complex parameters and $\mathcal{D}$ is a polynomial in the $\mathbf{z} = (z_1, \dots, z_n)$ with coefficients $\mathbf{x} = (x_1, \dots, x_m)$ [5]. In general, GKZ systems can be constructed for denominators which are a product of different polynomials, $\prod_j \mathcal{D}_j$ [9], but we do not require this here.

We define the polynomial $\mathcal{D}$ as

$$\mathcal{D} = \sum_m x_m \prod_{i=1}^n z_i^{(a_m)_i}, \quad (2.2)$$

where the $a_m \in \{a\}$ are n -dimensional column vectors containing the exponents of each of the $z_i \in \mathbf{z}$ in the m -th term of $\mathcal{D}$. The notation $(a_m)_i$ means selecting the i -th component of a_m . For example, the polynomial

$$\mathcal{D} = x_1 z_1 + x_2 z_2 + x_3 z_1 z_2 + x_4 \quad (2.3)$$

has vectors $a_1 = (1, 0)^T$, $a_2 = (0, 1)^T$, $a_3 = (1, 1)^T$, and $a_4 = (0, 0)^T$. In the GKZ formalism, the coefficients $\mathbf{x}$ become the variables that the integral depends upon, and the differential equations solved by it are differential equations in these variables [9, 10].

We can bring together the vectors a_m by interpreting them as column vectors of a matrix A , which we call the A -matrix of $\mathcal{D}$ or of the GKZ system; these terms are interchangeable as they both fully describe the GKZ system:

$$A = \begin{bmatrix} a_1 & \cdots & a_m \end{bmatrix}. \quad (2.4)$$

The A -matrix has, by construction, m columns, one for each term in $\mathcal{D}$, and n rows, for each $z_i \in \mathbf{z}$, $1 \leq i \leq n$. For instance, the GKZ system of Equation 2.3 has the corresponding A -matrix

$$A = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 \end{bmatrix} \quad (2.5)$$

Lastly, we introduce the augmented A -matrix, denoted $\mathcal{A}$, which we define as

$$\mathcal{A} = \begin{bmatrix} \mathbf{1} \\ A \end{bmatrix}, \quad (2.6)$$

where $\mathbf{1}$ is an m -dimensional row vector of ones. Augmenting the A -matrix in this way makes sense once we discuss the differential equations satisfied by the GKZ system. In our context, where we consider only one $\mathcal{D}$, both A and $\mathcal{A}$ fully describe the GKZ system, and so can be used interchangeably, but this is not true in the general case.

2.2 Differential equations

Associated with GKZ systems are two different sets of differential equations: the Euler and toric equations. The differential equations are obeyed by the GKZ integral, Equation 2.1. Here, we will show the general properties of the differential equations and explain why they work in general, but omit detailed derivations. For more in-depth reviews of these differential equations, see [8, 9].

Here and throughout, we use the convention for partial derivatives

$$\partial_i := \frac{\partial}{\partial x_i}, \quad (2.7)$$

and for the homogeneous differential operators,

$$\theta_i := x_i \partial_i. \quad (2.8)$$

2.2.1 Euler equations The Euler equations arise from integrating I by parts, assuming boundary terms vanish. For instance, consider integrating I by parts with respect to z_1 :

$$\begin{aligned} \int_0^\infty dz_1 \frac{\partial}{\partial z_1} \left(z_1^{\gamma_1} \left(\prod_{i \neq 1}^n \int_0^\infty dz_i z_i^{\gamma_i - 1} \right) \mathcal{D}^{-\gamma_0} \right) &= 0 \\ \gamma_1 I + \left(\prod_{i \neq 1}^n \int_0^\infty dz_i z_i^{\gamma_i - 1} \right) z_1 \frac{\partial}{\partial z_1} \mathcal{D}^{-\gamma_0} &= 0. \end{aligned} \quad (2.9)$$

Focusing on the last term of the above expression, we consider what happens when we operate with $z_1 \frac{\partial}{\partial z_1}$ on $\mathcal{D}^{-\gamma_0}$. To illustrate this, we use our example polynomial, Equation 2.3, and we have:

$$\begin{aligned} z_1 \frac{\partial}{\partial z_1} \mathcal{D}^{-\gamma_0} &= z_1 \frac{\partial}{\partial z_1} \left((x_1 z_1 + x_2 z_2 + x_3 z_1 z_2 + x_4)^{-\gamma_0} \right) \\ &= -\gamma_0 z_1 (x_1 + x_3 z_2) (x_1 z_1 + x_2 z_2 + x_3 z_1 z_2 + x_4)^{-\gamma_0-1} \\ z_1 \frac{\partial}{\partial z_1} \mathcal{D}^{-\gamma_0} &= -\gamma_0 z_1 (x_1 + x_3 z_2) \mathcal{D}^{-\gamma_0-1}. \end{aligned} \quad (2.10)$$

We notice that this result is equivalent to operating on $\mathcal{D}^{-\gamma_0}$ with the homogeneous differential operators θ_1 and θ_3 :

$$\begin{aligned} \theta_1 \mathcal{D}^{-\gamma_0} &= -\gamma_0 x_1 z_1 \mathcal{D}^{-\gamma_0-1}, \\ \theta_3 \mathcal{D}^{-\gamma_0} &= -\gamma_0 x_3 z_1 z_2 \mathcal{D}^{-\gamma_0-1}, \end{aligned} \quad (2.11)$$

where we recognise that by acting with $\theta_1 + \theta_3$, we get

$$(\theta_1 + \theta_3) \mathcal{D}^{-\gamma_0} = -\gamma_0 z_1 (x_1 + x_3 z_2) \mathcal{D}^{-\gamma_0-1} = z_1 \frac{\partial}{\partial z_1} \mathcal{D}^{-\gamma_0}. \quad (2.12)$$

By substituting the above into the integration by parts, we can take the homogeneous differential operators outside the integral, and obtain the expression

$$(\gamma_1 + \theta_1 + \theta_3) I = 0. \quad (2.13)$$

Repeating this for z_2 , and in general for the remaining z_i , yields the set of Euler equations. The differential operators in the Euler equations are known as the Euler operators, for example, $(\theta_1 + \theta_3)$. We can obtain a final identity, known as a Dilation Ward Identity (DWI) [5], recognising that

$$\left(\gamma_0 + \sum_{j=1}^n \theta_j \right) I = 0. \quad (2.14)$$

The Euler equations can in fact be obtained by reading off the rows of the $\mathcal{A}$ -matrix of the GKZ system. To illustrate this, we can turn the Euler equations into a matrix problem by considering a vector of the homogeneous differential operators Θ , which we define as

$$\Theta := (\theta_i), \quad 1 \leq i \leq n. \quad (2.15)$$

Then, applying Θ to the $\mathcal{A}$ -matrix, we get a new vector $\mathcal{E}$:

$$\mathcal{E} := \mathcal{A} \cdot \Theta, \quad (2.16)$$

where $\cdot$ represents the vector dot product [9]. The vector $\mathcal{E}$ contains the full set of Euler operators of a given $\mathcal{A}$ -matrix. The Euler equations can therefore be written in the form

$$(\gamma + \mathcal{E}) I = 0. \quad (2.17)$$

The DWI justifies augmenting the $\mathcal{A}$ -matrix with the row of ones, as it integrates this important identity seamlessly within the GKZ system.

2.2.2 Toric equations The full derivation of the toric equations requires a grasp of toric ideals. Nonetheless, we can understand them by once again considering our example GKZ system, denominated given in Equation 2.3.

We notice that the denominator in the GKZ integral, $\mathcal{D}^{-\gamma_0}$, when acted upon by certain differential operators in succession, yields the same result. For our example, consider acting with $\partial_1 \partial_2$

$$\partial_1 \partial_2 \mathcal{D}^{-\gamma_0} = \gamma_0 (\gamma_0 + 1) z_1 z_2 \mathcal{D}^{-\gamma_0-2} \quad (2.18)$$

and with $\partial_3 \partial_4$,

$$\partial_3 \partial_4 \mathcal{D}^{-\gamma_0} = \gamma_0 (\gamma_0 + 1) z_1 z_2 \mathcal{D}^{-\gamma_0-2}. \quad (2.19)$$

As we can see, acting on the denominator with $\partial_1\partial_2$ and $\partial_3\partial_4$ is the same. If we now consider the action of these operators on the integral, the result is the same, so equating the expressions, we get a toric equation:

$$(\partial_1\partial_2 - \partial_3\partial_4) I = 0 \quad (2.20)$$

In this example, only one toric equation exists. Unlike the Euler equations, the process of finding the toric equations satisfied by a GKZ system is not trivial. In general, it is a computational problem, involving finding the kernel of the $\mathcal{A}$ -matrix, $\ker \mathcal{A}$, and then finding the Gröbner basis of $\ker \mathcal{A}$ [8].

2.3 Polytopes

We now introduce the concept of a Newton polytope associated with GKZ systems. We define a Newton polytope, denoted Δ_A , as the convex hull of the set of vectors $\{a\}$, which are the columns of the A -matrix. In other words, we interpret each column $a_i \in \{a\}$ as points in n -dimensional parameter space; the convex hull of the set $\{a\}$ is the smallest convex set that completely encloses that set (convex meaning that we can draw a line between two points in the set such that all points on the line are in the set [11]). We denote this set as

$$\Delta_A := \text{conv}\{a\}, \quad (2.21)$$

and when written in this way, the polytope is in its vertex form. Polytopes can also be uniquely defined as an intersection of a finite number of half planes. It is always possible to go from a vertex-form to a half-space-intersection form [10, 12]; this requires a computational algorithm, many of which are widely implemented in standard packages such as SciPy [13]. We will not need to use this latter form, given that we always have access to the vertex form, given by the A -matrix.

Importantly, these polytopes live in the parameter space γ . Moving along the i -th direction in the parameter space corresponds to changing the value of the parameter γ_i . Meanwhile, γ_0 acts as a measure for the space, determining the spacing between lattice points. The volume of the polytope represents the values of γ for which the GKZ integral I is finite.

In the case of our example A -matrix, Equation 2.5, we can plot the polytope relatively simply:

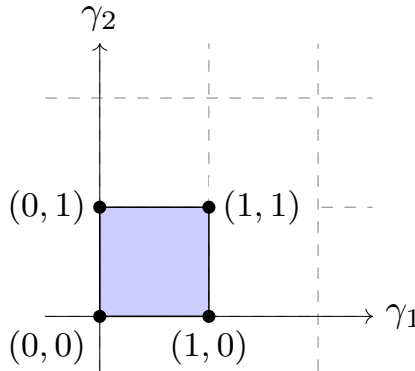


Figure 1: The lattice polytope for Equation 2.3. The polytope lives in the parameter space of γ . By convention, we drop the gammas in labelling the points; for example, we write the point (γ_1, γ_2) as $(1, 1)$.

As we can see from Figure 1, our example polynomial results in a polytope which is equivalent to the unit square. In this case, as the polynomial is only in two variables, z_1 and z_2 , the polytope lives in a two-dimensional space. Importantly, the polytope is a *lattice* polytope, meaning that its vertices lie on an integer lattice. This is a strict requirement set by the underlying theory of GKZ, which comes from the fact that the polytope represents the exponents in polynomial $\mathcal{D}$, which are, by definition, integer numbers.

2.4 Affine transformations

Consider taking a linear combination of a set of Euler equations, to form a new set of differential equations. At the same time, we could add to each Euler equation some multiple of the DWI, which would correspond to left-multiplying the $\mathcal{A}$ -matrix by an $(n+1) \times (n+1)$ matrix $\mathcal{M}$ of the form

$$\mathcal{M} = \begin{bmatrix} 1 & \mathbf{0} \\ \mathbf{b} & M \end{bmatrix}, \quad (2.22)$$

where $\mathbf{0}$ is an n -dimensional row vector of zeroes, $\mathbf{b}$ is an n -dimensional column vector, and M is an $n \times n$ matrix. Applied to $\mathcal{A}$, we have

$$\mathcal{A}' = \mathcal{M}\mathcal{A} = \begin{bmatrix} 1 & \mathbf{1} \\ \mathbf{b} & M \end{bmatrix} \begin{bmatrix} \mathbf{1} \\ A \end{bmatrix} = \begin{bmatrix} \mathbf{1} \\ A' \end{bmatrix} \quad (2.23)$$

where the components of A are affinely transformed to A' by

$$(A')_{ij} = a'_{ij} = b_i + \sum_{k=1}^n m_{ik} a_{kj}. \quad (2.24)$$

Concurrently, from the columns of the M -matrix, we can read off the appropriate change of variables to get from one integral form to the other. In the same way that the columns of the A -matrix encodes the power of each z_i in each term in $\mathcal{D}$, the columns of the M -matrix describes the power of each z_i in the change of variable $\mathbf{z} \rightarrow \mathbf{z}'$.

For instance, take

$$M = \begin{bmatrix} 0 & -1 & \frac{1}{2} \\ -1 & 0 & 1 \\ \frac{1}{2} & 1 & 0 \end{bmatrix}. \quad (2.25)$$

If we augment the matrix to show the transformed variables z'_i and original variables z_i , we can write down the change of variables:

	z'_1	z'_2	z'_3
z_1	0	-1	$\frac{1}{2}$
z_2	-1	0	1
z_3	$\frac{1}{2}$	1	0

(2.26)

Then it is easy to see that the equivalent change of variables $\mathbf{z} \rightarrow \mathbf{z}'$, in this case, is

$$z'_1 = z_2^{-1} z_3^{\frac{1}{2}}, \quad z'_2 = z_1^{-1} z_3, \quad z'_3 = z_1^{\frac{1}{2}} z_2. \quad (2.27)$$

Conversely, knowing the change of variables $\mathbf{z} \rightarrow \mathbf{z}'$ allows us to construct the M -matrix. Then, to get the $\mathcal{M}$ -matrix requires simple algebra to find the $\mathbf{b}$ vector.

Consider performing the change of variables corresponding to an affine transformation represented by a matrix $\mathcal{M}$. The new matrix A' defines a new set of Euler equations,

$$(\gamma' + \mathcal{E}')I = 0 \quad (2.28)$$

where $\mathcal{E}' = \mathcal{A}' \cdot \Theta$ and the parameter vector also changes according to

$$\gamma' = \begin{bmatrix} 1 & \mathbf{0} \\ \mathbf{b} & M \end{bmatrix} \gamma = \mathcal{M}\gamma \quad (2.29)$$

Note that the transformation does not affect the DWI. The toric equations are unaffected by an affine transformation, provided that $\det(\mathcal{M}) \neq 0$, as left-multiplying by an invertible matrix preserves the kernel of the matrix [14].

On the integral side, performing the change of variables provides us with a crucial relation between the GKZ integrals themselves. It can be shown that

$$I' = \frac{I}{|\det \mathcal{M}|}, \quad (2.30)$$

meaning that if the $\mathcal{A}$ -matrices of two GKZ systems are related by an affine transformation, then their respective GKZ integrals are proportional to each other, and the solution is effectively unchanged up to a constant prefactor [6].

3 Connection to Feynman diagrams

In general, Feynman diagrams and their associated integrals represent the probability amplitude of that interaction occurring [15].

So far, we have discussed GKZ integrals, but we do not yet understand how they connect to the core of this dissertation, the Feynman integrals. In order to understand this link, we first discuss the different representations of Feynman diagrams as integrals, specifically in the LP representation.

Throughout this section and beyond, we will neglect multiplicative prefactors of Feynman integrals, given that they become redundant once we reach GKZ form. Therefore, all integrals are presented as being equal *up to a prefactor* to focus on what is inside the integral.

3.1 Momentum and position representation

The simplest representation of a Feynman diagram is in the momentum representation. In this representation, we assign to each edge in the diagram an inverse propagator. Then, for each loop, we integrate the product of the inverse propagators over all the independent loop momenta k_i .

Specifically, the inverse propagators are defined such that

$$\frac{q, m, \nu}{\rightarrow} = \frac{1}{(-q^2 + m^2)^\nu}, \quad (3.1)$$

where q is a momentum, m a mass, and ν the order to which the propagator appears. For each loop in the diagram, we can associate to it a loop momentum k_i , effectively parametrisng the momenta by invoking conservation of momentum at each step. We define a Feynman diagram with l loops and n internal edges in D space-time dimensions as

$$I = \int \prod_{r=1}^l d^D k_r \prod_{i=1}^n \frac{1}{(-q_i^2 + m_i^2)^{\nu_i}}. \quad (3.2)$$

3.2 LP representation

The LP representation reveals the connection to the GKZ integrals as discussed in § 2. We start with Equation 3.2, in the momentum-space representation, and show how it is parameterised into the LP representation.

We begin by performing a Wick rotation, which brings the integral out of momentum space, into position space. The integral of Equation 3.2 comes with a Minkowskian signature, but it is easier to work with in a Euclidean one. The effect of the Wick rotation is to change the sign of the squared momenta, up to a prefactor, such that $-q_i^2 \rightarrow +q_i^2$, and the integral is then

$$I = \int \prod_{r=1}^l d^D k_r \prod_{i=1}^n \frac{1}{(q_i^2 + m_i^2)^{\nu_i}}. \quad (3.3)$$

Now, we use Schwinger's trick [15], which reformulates each of the individual inverse operators as a product of integrals of exponentials over n Schwinger parameters. From the definition of the gamma function $\Gamma(\nu)$, we have the identity

$$\frac{1}{A^\nu} = \frac{1}{\Gamma(\nu)} \int_0^\infty dt_i t_i^{\nu-1} e^{-t_i A}. \quad (3.4)$$

Applying this to each inverse propagator in I , we obtain

$$\begin{aligned} I &= \int \prod_{r=1}^l d^D k_r \prod_{i=1}^n \int_0^\infty dt_i t_i^{\nu_i-1} e^{-t_i (q_i^2 + m_i^2)} \\ &= \int d^n t \left(\prod_{i=1}^n t_i^{\nu_i-1} \right) \prod_{r=1}^l \int_0^\infty d^D k_r \exp \left(- \sum_{i=1}^n t_i (q_i^2 + m_i^2) \right). \end{aligned} \quad (3.5)$$

From here, it is generally possible to complete the square in the exponent, integrate over the k_r , to obtain the integral in its Schwinger representation:

$$I = \int_0^\infty dt \left(\prod_{i=1}^n t_i^{\nu_i-1} \right) (\mathcal{U}(t))^{-\frac{D}{2}} \exp \left(- \frac{\mathcal{F}(t)}{\mathcal{U}(t)} \right), \quad (3.6)$$

where $\mathcal{U}$ and $\mathcal{F}$ are known as the First and Second Symanzik polynomials, respectively. Following Weinzierl [15], we can rewrite the Schwinger integral into the LP form to get

$$I = \int_0^\infty d^n t \left(\prod_{i=1}^n t_i^{\nu_i-1} \right) (\mathcal{G}(t))^{-\frac{D}{2}}, \quad (3.7)$$

where $\mathcal{G} = \mathcal{U} + \mathcal{F}$. At this point, the Feynman integral is of the same form as that of the GKZ integral of Equation 2.1. In order to get the GKZ system, we say that the coefficients of the t_i in $\mathcal{G}$ become indeterminate, a process that we call *promotion*. Conversely, we may *restrict* the indeterminate coefficients of a GKZ system to get back to a Feynman or other known integral.

4 The Triangle Feynman diagram

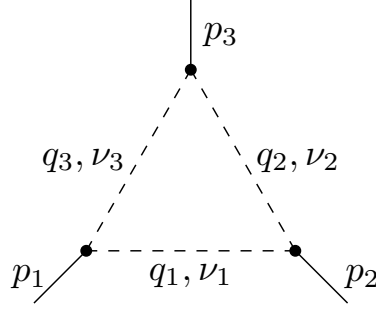


Figure 2: The massless triangle Feynman diagram, where ν_i are the orders of the propagators, q_i and p_i represent the loop momenta and external momenta, respectively. At each vertex in the diagram, we enforce conservation of momentum.

The triangle Feynman diagram, pictured in Figure 2, is a well-known diagram used to describe the Casimir effect, Hawking radiation, and Lamb shift. It has been well-studied and has analytical solutions [16–18]. In terms of inverse propagators, in the massless case, it can be written as

$$I_{\text{TR}} = \int_0^\infty d^D \mathbf{k} \frac{1}{-q_1^{2\nu_1}} \frac{1}{-q_2^{2\nu_2}} \frac{1}{-q_3^{2\nu_3}}. \quad (4.1)$$

We apply a Wick rotation ($-q_i \rightarrow +q_i$) to move the integrand from Minkowskian to Euclidean signature, and constrain each q_i to a single loop momentum k by momentum conservation. Specifically,

$$q_1 \rightarrow k, \quad q_2 \rightarrow k - p_1, \quad q_3 \rightarrow k + p_2. \quad (4.2)$$

Substituting into the integral, we have

$$I_{\text{TR}} = \int_0^\infty d^D k \frac{1}{k^{2\nu_1}} \frac{1}{(k - p_1)^{2\nu_2}} \frac{1}{(k + p_2)^{2\nu_3}}. \quad (4.3)$$

Applying Schwinger's trick, Equation 3.4, to each inverse propagator,

$$I_{\text{TR}} = \int_0^\infty d^D k \left(\prod_{i=1}^3 \int_0^\infty dt_i t_i^{\nu_i-1} \right) \exp(-t_1 k^2) \exp(-t_2 (k - p_1)^2) \exp(-t_3 (k + p_2)^2). \quad (4.4)$$

In order to get this integral into the LP representation, we first complete the square for k in the exponential. We have

$$t_1 k^2 + t_2 (k - p_1)^2 + t_3 (k + p_2)^2 = (t_1 + t_2 + t_3) \left(k - \frac{p_1 t_2 - p_2 t_3}{t_1 + t_2 + t_3} \right)^2 + \frac{p_1^2 t_1 t_2 + p_2^2 t_1 t_3 + p_3^2 t_2 t_3}{t_1 + t_2 + t_3}, \quad (4.5)$$

where $p_3 = p_1 + p_2$, from momentum conservation. Now, we can integrate over k . We observe that the only k -term in the integrand comes from the first term and is in the form of a Gaussian integral, so

$$\int_0^\infty d^D k \exp\left(-(t_1 + t_2 + t_3)\left(k - \frac{p_1 t_2 - p_2 t_3}{t_1 + t_2 + t_3}\right)\right) = \int_0^\infty d^D \kappa e^{-\tau \kappa^2} = \tau^{-\frac{D}{2}}, \quad (4.6)$$

up to a constant prefactor [19], where $\tau = t_1 + t_2 + t_3$ and $\kappa = k - \frac{p_1 t_2 - p_2 t_3}{\tau}$. Therefore, Equation 4.4 becomes

$$I_{\text{TR}} = \left(\prod_{i=1}^3 \int_0^\infty dt_i t_i^{\nu_i-1}\right) (t_1 + t_2 + t_3)^{-\frac{D}{2}} \exp\left(\frac{p_1^2 t_2 t_3 + p_2^2 t_1 t_3 + p_3^2 t_1 t_2}{t_1 + t_2 + t_3}\right), \quad (4.7)$$

where we can recognise this as being in the Schwinger representation [15], with the Symanzik polynomials $\mathcal{U} = t_1 + t_2 + t_3$ and $\mathcal{F} = p_1^2 t_1 t_2 + p_2^2 t_1 t_3 + p_3^2 t_2 t_3$. Converting to the LP representation, this yields

$$I_{\text{TR}} = \left(\prod_{i=1}^3 \int_0^\infty dt_i t_i^{\nu_i-1}\right) \mathcal{G}^{-\gamma_0}, \quad (4.8)$$

with $\mathcal{G} = \mathcal{F} + \mathcal{U} = p_1^2 t_2 t_3 + p_2^2 t_1 t_3 + p_3^2 t_1 t_2 + t_1 + t_2 + t_3$. Finally, to arrive at the GKZ representation of the triangle integral, we must promote the coefficients of the t_i terms in $\mathcal{G}$ to be indeterminate:

$$\mathbf{x} = (p_1^2, p_2^2, p_3^2, 1, 1, 1), \quad (4.9)$$

and so the GKZ integral can be written as

$$I_{\text{TR}} = \frac{\prod_{i=1}^3 \int dt_i t_i^{\gamma_i-1}}{(x_1 t_2 t_3 + x_2 t_1 t_3 + x_3 t_1 t_2 + x_4 t_1 + x_5 t_2 + x_6 t_3)^{\gamma_0}}. \quad (4.10)$$

Finally, from the denominator of I_{TR} , we get the triangle's $\mathcal{A}$ -matrix,

$$\mathcal{A}_{\text{TR}} = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 & 0 & 1 \end{bmatrix}. \quad (4.11)$$

5 The Triple K

The triple K (3K) integral is a representation of a three-point correlation function and appears in conformal field theory and cosmology [5, 20]. The 3K integral itself can be written as

$$I_{3K} = \int_0^\infty dz z^\alpha \prod_{i=1}^3 p_i^{\beta_i} K_{\beta_i}(p_i z) = \int_0^\infty dz z^\alpha p_1^{\beta_1} K_{\beta_1}(p_1 z) p_2^{\beta_2} K_{\beta_2}(p_2 z) p_3^{\beta_3} K_{\beta_3}(p_3 z) \quad (5.1)$$

where $\alpha = \frac{D}{2} - 1$ for the triple K, and $K_{\beta_i}(z)$ is the modified Bessel function of the second kind [21], defined as

$$K_{\beta_i}(z) = \frac{1}{2} \left(\frac{z}{2}\right)^{\beta_i} \int_0^\infty dt_i t_i^{-\beta_i-1} e^{-t_i - \frac{z^2}{4t_i}}, \quad (5.2)$$

see Equation 10.32.10 in [22].

In order to bring I_{3K} to GKZ form as in Equation 2.1, we begin by working on a single $K_{\beta_i}(p_i z)$ function. We begin by performing the change of variables $t_i \rightarrow T_i$:

$$t_i = \frac{p_i^2 z}{2T_i}, \quad dt_i = -\frac{p_i^2 z}{2T_i^2} dT_i \quad (5.3)$$

where $\lim_{t_i \rightarrow 0}(T_i) = +\infty$ and $\lim_{t_i \rightarrow +\infty}(T_i) = 0$.

Substituting Equation 5.3 into Equation 5.2 with the appropriate limits, we have

$$\begin{aligned}
K_{\beta_i}(p_i z) &= \frac{1}{2} \left(\frac{p_i z}{2} \right)^{\beta_i} \int_{-\infty}^0 dT_i \left(\frac{p_i^2 z}{2T_i} \right)^{-\beta_i-1} \left(-\frac{p_i^2 z}{2T_i^2} \right) \exp \left(-\left(\frac{p_i^2 z}{2T_i} \right) + \frac{p_i^2 z}{4 \left(\frac{p_i^2 z}{2T_i} \right)} \right) \\
&= \frac{1}{2} p_i^{-\beta_i} \int_0^{\infty} dT_i T_i^{\beta_i-1} \exp \left(-\left(\frac{p_i^2 z}{2T_i} + \frac{z}{2} T_i \right) \right) \\
p_i^{\beta_i} K_{\beta_i}(p_i z) &= \frac{1}{2} \int_0^{\infty} dT_i T_i^{\beta_i-1} \exp \left(-\frac{z}{2} \left(\frac{p_i^2}{T_i} + T_i \right) \right)
\end{aligned} \tag{5.4}$$

where the limits of integration are swapped by factoring the negative sign from the Jacobian. Substituting Equation 5.4 into Equation 5.1, ignoring prefactors, and performing the z -integration, we obtain

$$I_{3K} = \int_0^{\infty} dz z^{\alpha} \left(\prod_{i=1}^3 \int_0^{\infty} dT_i T_i^{\gamma_i-1} \right) \left(\sum_{i=1}^3 \left(\frac{p_i^2}{T_i} + T_i \right) \right)^{-\gamma_0}, \tag{5.5}$$

where we define $\gamma_0 = \alpha + 1$ [6] and renaming the $\beta_i \rightarrow \gamma_i$. Finally, we promote the coefficients of the monomials in the T_i 's to get the GKZ system for the triple K:

$$I_{3K} = \left(\prod_{i=1}^3 \int_0^{\infty} dT_i T_i^{\gamma_i-1} \right) \mathcal{G}^{-\gamma_0}, \tag{5.6}$$

where

$$\mathcal{G} = \frac{x_1}{T_1} + \frac{x_2}{T_2} + \frac{x_3}{T_3} + x_4 T_1 + x_5 T_2 + x_6 T_3. \tag{5.7}$$

The original triple K integral corresponds to the restriction of Equation 5.6 with

$$\gamma_i = \beta_i, \quad \gamma_0 = \alpha + 1, \quad \mathbf{x} = (p_1^2, p_2^2, p_3^2, 1, 1, 1). \tag{5.8}$$

From Equation 5.7, we see that the $\mathcal{A}$ -matrix of the triple K matrix is

$$\mathcal{A}_{3K} = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 \\ -1 & 0 & 0 & 1 & 0 & 0 \\ 0 & -1 & 0 & 0 & 1 & 0 \\ 0 & 0 & -1 & 0 & 0 & 1 \end{bmatrix}. \tag{5.9}$$

Moreover, it is trivial to show that, from Equation 5.4, a product of N -K functions in an integral of the form

$$I_{NK} = \int_0^{\infty} dz z^{\alpha} \left(\prod_{i=1}^N p_i^{\beta_i} K_{\beta_i}(p_i z) \right), \tag{5.10}$$

has an A -matrix of the form

$$A_{NK} = \left[\begin{array}{c|c} -\mathbb{I}_N & \mathbb{I}_N \end{array} \right], \tag{5.11}$$

where $\mathbb{I}_N$ is the N -dimensional identity matrix. In effect, the N -K integral's polytope is the lattice simplex polytope in N -dimensional parameter space, symmetric about the origin.

6 Transformations between Feynman diagrams

We now investigate the relationships between pairs of Feynman diagrams, now that we understand how they can be written in the form of a GKZ system. Equipped with the knowledge that two GKZ systems are equivalent if they are related by an affine transformation as in Equation 2.22, we begin investigating the necessary conditions for which two GKZ systems are equivalent.

6.1 Properties of affine transformations

When we speak about transforming between different GKZ systems, we specifically mean applying affine transformations on their A -matrices, which can in turn be represented by their associated polytopes. We will focus on the polytopes, examining exactly how they change under the influence of the affine transformations, to see what information is preserved.

To this point, the primary feature of affine transformation is the fact that they preserve colinearity, in other words, parallelism [23]. As such, we would expect that the parallelism of all facets of the polytopes under examination is conserved under an affine transformation. Going further, this would suggest that the underlying *structure* of the polytope is conserved, but what this means at this point is abstract.

6.2 Triangle and Triple K

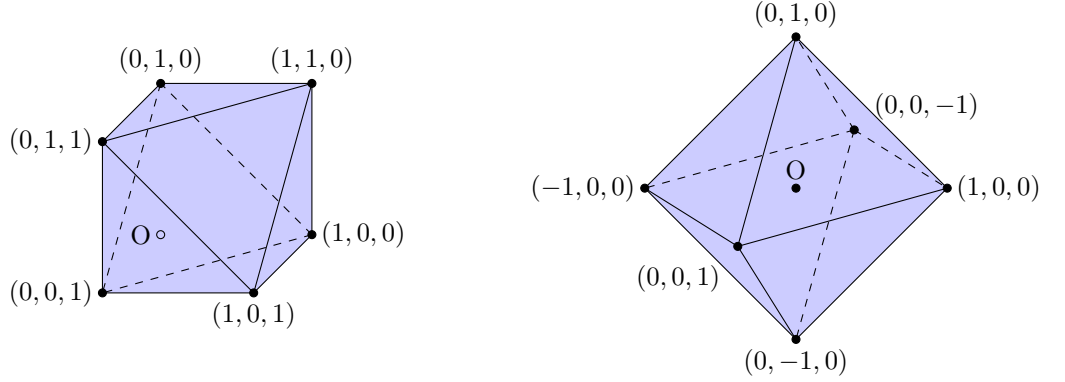


Figure 3: Polytopes representing the triangle integral, left, given by Equation 4.1, and the triple K, right, given by Equation 5.1, both drawn to scale. The origin, O , is also shown. Shaded in points represent points in the polytope, i.e. within the volume spanned by the points or on the boundary, while unshaded points are not within the polytope.

We start by looking at a worked example, the triangle and the triple K integrals. These are well-known in literature to be equivalent integrals, meaning that they should be related by an affine transformation.

Finding the $\mathcal{M}$ -matrix which describes the transformation from one to the other is a trivial problem which involves solving a system of simultaneous equations as in Equation 2.24. For the triangle-triple K, the transformation is described by the matrix

$$M_{3K \rightarrow TR} = \begin{bmatrix} \frac{1}{2} & -\frac{1}{2} & -\frac{1}{2} \\ -\frac{1}{2} & \frac{1}{2} & -\frac{1}{2} \\ -\frac{1}{2} & -\frac{1}{2} & \frac{1}{2} \end{bmatrix}, \quad \mathbf{b}_{3K \rightarrow TR} = \begin{bmatrix} \frac{1}{2} \\ \frac{1}{2} \\ \frac{1}{2} \end{bmatrix} \longrightarrow \mathcal{M}_{3K \rightarrow TR} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ \frac{1}{2} & \frac{1}{2} & -\frac{1}{2} & -\frac{1}{2} \\ \frac{1}{2} & -\frac{1}{2} & \frac{1}{2} & -\frac{1}{2} \\ \frac{1}{2} & -\frac{1}{2} & -\frac{1}{2} & \frac{1}{2} \end{bmatrix}. \quad (6.1)$$

Going the other way, from the triangle to the triple K, we find

$$M_{TR \rightarrow 3K} = \begin{bmatrix} 0 & -1 & -1 \\ -1 & 0 & -1 \\ -1 & -1 & 0 \end{bmatrix}, \quad \mathbf{b}_{TR \rightarrow 3K} = \begin{bmatrix} -\frac{1}{2} \\ -\frac{1}{2} \\ -\frac{1}{2} \end{bmatrix} \longrightarrow \mathcal{M}_{TR \rightarrow 3K} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 0 & -1 & -1 \\ 1 & -1 & 0 & -1 \\ 1 & -1 & -1 & 0 \end{bmatrix} \quad (6.2)$$

As we expected,

$$\mathcal{M}_{TR \rightarrow 3K} = \mathcal{M}_{3K \rightarrow TR}^{-1}. \quad (6.3)$$

We read off, from the columns of the M matrix, the appropriate change of variables required to move from one GKZ system to the other. For the triangle (z'_i variables) to the triple K (z_i variables), we have

$$z'_1 = \frac{1}{z_2 z_3}, \quad z'_2 = \frac{1}{z_1 z_3}, \quad z'_3 = \frac{1}{z_1 z_2}. \quad (6.4)$$

Both polytopes are octahedra, as can be seen in Figure 3, which confirms the assertion that affine transformations preserve parallelism of facets in polytopes. There is also a translation, as we can see that the origin is not in the volume of the triangle polytope, but is in the triple K's.

7 Extending to polytopes in different parameter-spatial dimensions

Having established the affine equivalence between the triangle and the triple K, we begin by looking for other cases where we hypothesised that the same correspondence between polytopes existed. In this section, we investigate whether the properties that arise from the affine equivalence of two GKZ systems holds more generally, beyond this specific example.

First, we focus on lower-dimensional examples. This way, we can visualise and graph the transformations more intuitively. However, as we will see, our investigation reveals oversights in our method, and exposes a deeper truth about these affine transformations.

7.1 Two dimensional example

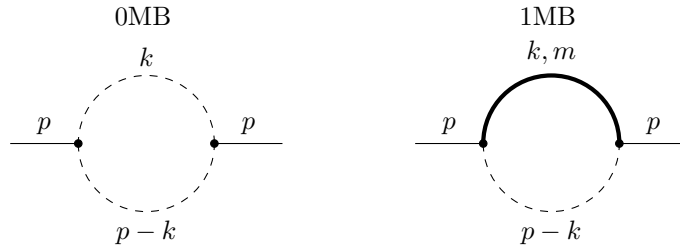


Figure 4: Feynman diagrams for the massless bubble (left), and the one-mass bubble (right).

We consider as an example the massless bubble integral (0MB), the one-mass bubble integral (1MB), and an integral we named the doubleK (2K). The choice of GKZ systems becomes clear when we plot their respective polytopes, but first we write down their $\mathcal{A}$ -matrices.

With the techniques used to derive the $\mathcal{A}$ -matrix of the triangle, we can derive the $\mathcal{A}$ matrices of the massless and one-mass bubbles. For the massless bubble:

$$\mathcal{A}_{0MB} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{bmatrix}, \quad (7.1)$$

where we adopt the convention of de la Cruz [10] and introduce a deformation to the Symanzik polynomials $r(z) = k$, namely we add a constant term to the Symanzik polynomials which is the $(1, 0, 0)^T$ column in the above $\mathcal{A}$ -matrix. To evaluate the integral, we take the limit as $k \rightarrow 0$. This procedure is advantageous as it ensures that the codimension of $\mathcal{A}_{0MB}$ is non-zero, so that $\ker \mathcal{A}_{0MB} \neq \emptyset$ and we can find the set of differential equations which this integral satisfies. As with the massless bubble, this Feynman diagram also has known analytical solutions [17, 18, 24]. This matrix is a permutation to our earlier example's $\mathcal{A}$ -matrix, Equation 2.5.

For the case of the one-mass bubble, the codimension of its $\mathcal{A}$ -matrix is non-zero, so we do not have to deform the Symanzik polynomials. This Feynman diagram also has known analytical solutions [18, 25]. Its $\mathcal{A}$ -matrix is

$$\mathcal{A}_{1MB} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 0 \\ 2 & 1 & 0 & 1 \end{bmatrix}. \quad (7.2)$$

Finally, we define the 2K integral as in Equation 5.10 with $N = 2$, and from Equation 5.11 we found the resulting $\mathcal{A}$ -matrix:

$$\mathcal{A}_{2K} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ -1 & 0 & 1 & 0 \\ 0 & -1 & 0 & 1 \end{bmatrix}. \quad (7.3)$$

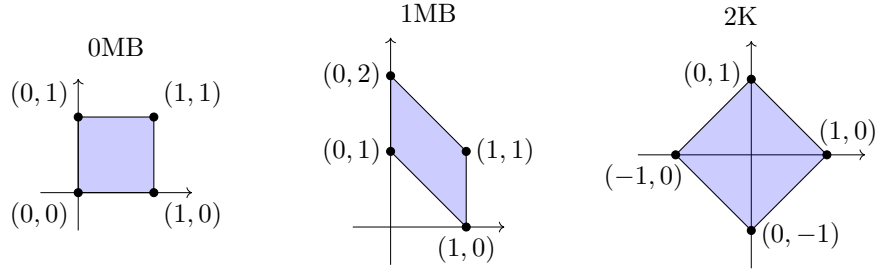


Figure 5: Polytopes for the massless bubble (left), one-mass bubble (centre), and 2K polytope (right).

We hypothesised that these GKZ systems were affinely equivalent to each other from knowing the shapes of their polytopes, Figure 5. It is possible to imagine affine transformations between pairs of the three polytopes above, for example, the 0MB to 1MB requires a shear, or the 2K to the 0MB requires a rotation, scaling and translation. In each case, the parallelism of the polytope is conserved; edges that start out as parallel in one polytope remain opposite and parallel in the transformed version.

So, in the same way as with the triangle and the 3K, we look for a matrix M and a vector $\mathbf{b}$ which described the transformation between two of these polytopes. Take, for instance, the affine transformation from the 2K to the massless bubble. We tried finding the transformation by solving for the components of M and $\mathbf{b}$:

$$\begin{bmatrix} 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{bmatrix} = \begin{bmatrix} m_{11} & m_{12} \\ m_{21} & m_{22} \end{bmatrix} \begin{bmatrix} -1 & 0 & 1 & 0 \\ 0 & -1 & 0 & 1 \end{bmatrix} + \mathbf{b} \cdot \mathbf{1}_{n \times m}, \quad (7.4)$$

where $\mathbf{1}_{n \times m}$ represents an n -by- m matrix full of ones.

Attempting to solve the system of simultaneous equations from the above matrix equation reveals that there is no solution. This implies that there is no affine transformation between these two polytopes, and hence that they are not equivalent. However, we based our hypothesis that they were equivalent on the conservation of the parallelism of the polytopes, demonstrated in Figure 5. Does this mean that the preservation of a polytope's structure is insufficient to guarantee affine equivalences?

7.2 The volume vector

In order to answer this question we look for a more rigorous test to determine whether the polytopes were affinely equivalent to begin with, and tell us whether a transformation of the type of Equation 2.24 is possible for two given polytopes. There are several different methods to this end, for example finding the labelled face graphs of the polytopes [26]. In fact, the problem of determining affine equivalence of two polytopes is still an active area of research in graph theory and computational geometry, and is considered graph isomorphism hard [27–29].

The approach that we chose came from a recent paper by Cai, Zhang, and Liu [30], which builds upon a theorem by Blanco and Santos [31]. In it, they prove that, under an affine transformation, a quantity known as the volume vector of a convex lattice polytope is conserved.

To understand the volume vector, consider a polytope Δ_A , which has an associated A -matrix, $A = [a_1 \cdots a_m]$. We define the volume vector of A as

$$w = [w_{1,\dots,n+1}], \quad |1, \dots, n+1| = \binom{m}{n+1}, \quad (7.5)$$

where $|\dots|$ denotes the number of elements in the list, and

$$w_{1,\dots,n+1} = \det \begin{pmatrix} 1 & \cdots & 1 \\ a_1 & \cdots & a_{n+1} \end{pmatrix}. \quad (7.6)$$

Suppose that $w = kw'$, $k \in \mathbb{Z}$, then we refer to w' as the primitive volume vector.

Computing the volume vector involves grouping the m points defining the polytope into all possible $d+1$ combinations, placing them in a matrix, which is by construction square, and taking its determinant. The components of w represents different combinations of the polytope's points.

To illustrate this procedure, we find the volume vector of our previous example, Equation 2.5. In this case, $m = 4$ and $n = 2$, so $\binom{m}{n+1} = 4$ and the volume vector will consist of four elements: the

four different ways of choosing three columns from the four columns of A . Applying the formula, we have

$$w = [w_{1,2,3}, w_{1,2,4}, w_{1,3,4}, w_{2,3,4}]. \quad (7.7)$$

The components of w can then be readily computed by taking determinants. If we take the first element, $w_{1,2,3}$, this refers to taking the determinant

$$w_{1,2,3} = \det \begin{pmatrix} 1 & 1 & 1 \\ a_1 & a_2 & a_3 \end{pmatrix} = \det \begin{pmatrix} 1 & 1 & 1 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix} = -1. \quad (7.8)$$

Doing the same for the rest of the components of w , we find that, for the matrix A given in Equation 2.5, the volume vector is

$$w = [-1, 1, 1, -1] \quad (7.9)$$

The theorem from Cai, Zhang, and Liu [30] states that “two polytopes P and Q are affinely equivalent if and only if P and Q have the same number of vertices, and after *proper ordering* of the vertices of P and Q , the primitive volume vectors of the vertex sets of P and Q are equal”.

Indeed, the key aspect that we had overlooked was this “proper ordering” of the columns of the A -matrix. The components of the volume vector are determinants of the columns of A , and we know that swapping columns in a determinant results in a change of sign. It is clear to see that, considering the above example, permuting the columns of A in some form would lead to taking different determinants, and therefore different volume vectors. In this sense, it is clear why ordering matters: the matrices of both polytopes must be ordered in the same fashion in order to ensure consistency when taking determinants of its columns.

There are multiple ways of ordering the columns of a polytope; as long as the ordering method is consistent across different polytopes, the volume vector test yields valid results. Since we originally chose an arbitrary ordering of the columns the outcome of that volume vector test is not valid.

The fact that ordering does matter is an interesting contradiction with the idea of GKZ, an issue we discuss later in § 9.1. We proceed with the assumption that the ordering of the columns matters, at least in determining the affine equivalence of two polytopes, which, ultimately, is our goal. There are several different methods of ordering columns of a matrix, but for now, we choose the simplest to implement, which is lexicographical ordering.

With this in mind, we tried sorting the columns of the A -matrices, and then finding an affine transformation. As we had expected, after proper ordering, the massless bubble, one-mass bubble, and the 2K all shared the same primitive volume vector, and hence were affinely equivalent. Moreover, we were able to find the matrix $\mathcal{M}$ which defined the affine transformation between each pair of polytopes. In the case of the 3K and triangle, the ordering that we chose so happened to be consistent across both, which made it possible to find the affine transformation.

7.3 Orderings

Our chosen ordering method was a simple lexicographic one. The specific lexicographical ordering was such that, given two column vectors $a_i, a_j \in A$, we say that

$$a_i <_{\text{lex}} a_j \quad (7.10)$$

if at the first index where they differ, a_i has a smaller value. Formally,

$$a_i <_{\text{lex}} a_j \quad \text{if } \exists \lambda : \begin{cases} (a_i)_k = (a_j)_k & \forall k < \lambda \\ (a_i)_\lambda < (a_j)_\lambda \end{cases} \quad (7.11)$$

We permute the columns of $A = [a_1, a_2, \dots, a_m]$ such that

$$a_1 <_{\text{lex}} a_2 <_{\text{lex}} \dots <_{\text{lex}} a_m. \quad (7.12)$$

At this point, we say that A is lexicographically ordered. When two A -matrices are ordered in this fashion, they yield accurate results when subjected to the volume vector test.

Some authors choose to use other methods to ensure that polytopes are compared correctly. For example, Grinis and Kasprzyk [26] describe how to find the so-called normal form of a polytope,

using the algorithm from the PALP package [32]. After implementing their algorithm, we find that while we attain the expected results in terms of affine equivalence, the normal forms of the polytopes do not yield information about the affine transformation which transforms one into the other. For this reason, we believe that simply ordering the columns of A in some canonical fashion is far more powerful in our use case.

7.4 Mapping point transformations

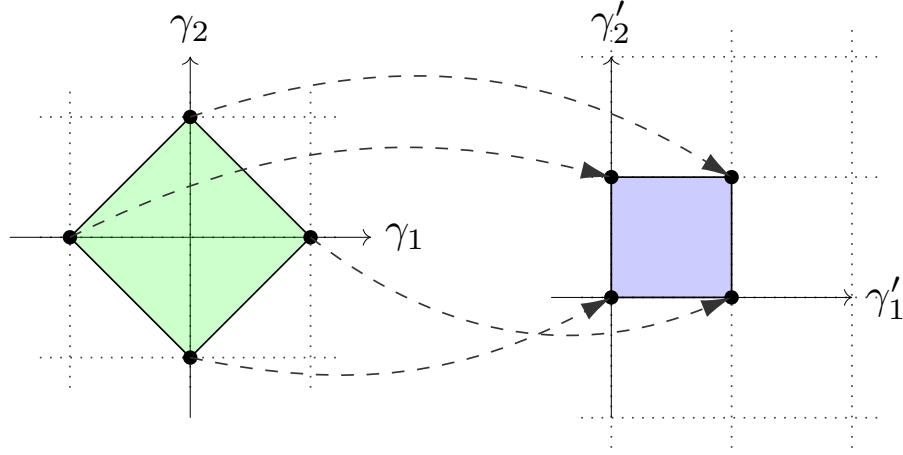


Figure 6: Mapping each vertex of the 2K (left) to the vertices of the 0MB (right) under an affine transformation $\mathcal{M}$. Dashed lines shows corresponding columns in their respective A -matrices.

With the A -matrices properly ordered, we can now see how the vertices of the polytope move under an affine transformation. The matrix-multiplication acts column-wise, so on individual column vectors (points in parameter space) a_i are transformed into new column vectors a'_i by

$$a'_i = Ma_i + \mathbf{b}. \quad (7.13)$$

This means that we can effectively track the how each point in one polytope transforms into the new polytope. We would expect that the combinatorics of the polytope to remain the same, that is, the relationship between the facets is the same, given the properties of an affine transformation. However, the mapping of each point to a new point, in terms of the kinematic properties that each point represents, is not trivial.

In the case of the 2K and 0MB, shown in Figure 6, the mapping of each vertex of the polytope is clear. We can notice that this particular M -matrix applies a rotation by $\frac{\pi}{4}$ clockwise, and an appropriate scaling and translation to ensure that the resulting polytope lies on the lattice. Importantly, we observe that the affine transformation preserves the combinatorial structure of the polytope: vertices that originally lay on a facet are mapped to a corresponding facet under the affine transformation.

7.5 Four dimensional example

Equipped with new tools for establishing affine equivalence, we apply them to four-dimensional polytopes.

As our base, we start with the polytope of a GKZ system comprising of a product of four modified Bessel functions, also known as the four K (4K) integral. As the 4K is the most symmetric lattice simplex, we consider it the most natural starting point. Actually visualising these polytopes is impossible, as they live in a higher-dimensional parameter space, so we rely entirely on our volume vector to test equivalence, rather than on shape, as before. The 4K polytope, according to Equation 5.11 with $N = 4$, has the A -matrix

$$A_{4K} = \left[\begin{array}{c|c} \mathbb{I}_4 & -\mathbb{I}_4 \end{array} \right], \quad (7.14)$$

where $\mathbb{I}_4$ represents the four-dimensional identity matrix, so A_{4K} has dimensions four-by-eight.

We are unable to find any known Feynman diagrams or integrals with GKZ systems affinely equivalent to A_{4K} . So, instead, we construct an affinely equivalent polytope, by creating an affine transformation matrix $\mathcal{M}$. When applied to A_{4K} , it creates an affinely equivalent polytope and A -matrix, by definition.

In this case, the matrix-side of the affine transformations is fully understood, given that we have written out the transformation ourselves. However, we want to see how the change of variables side of the transformation works. The affine-equivalent candidate polytope (can.) to the 4K that we found has the A -matrix

$$A_{\text{can.}} = \begin{bmatrix} 1 & 0 & 1 & 1 & 1 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 & 0 & 1 & 0 & 1 \\ 1 & 0 & 0 & 0 & 1 & 0 & 1 & 1 \end{bmatrix}. \quad (7.15)$$

The matrix M and vector $\mathbf{b}$ relating these two polytopes are

$$M_{4K \rightarrow \text{can.}} = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} & -\frac{1}{2} & -\frac{1}{2} \\ \frac{1}{2} & -\frac{1}{2} & \frac{1}{2} & -\frac{1}{2} \\ \frac{1}{2} & -\frac{1}{2} & -\frac{1}{2} & \frac{1}{2} \end{bmatrix}, \quad \mathbf{b}_{4K \rightarrow \text{can.}} = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}. \quad (7.16)$$

Reading off the columns of the above M matrix gives us the appropriate change of variables, giving us the transformed variables $\mathbf{z}'$ as a function of the original variables $\mathbf{z}$. We need the transformation $\mathbf{z} \rightarrow \mathbf{z}'$, so we invert the matrix $M_{4K \rightarrow \text{can.}}$. In fact, $M_{4K \rightarrow \text{can.}}$ matrix is involutory, $M_{4K \rightarrow \text{can.}} = M_{\text{can.} \rightarrow 4K}$, and so we can write the change of variables as

$$z_1 = \sqrt{z'_1 z'_2 z'_3 z'_4}, \quad z_2 = \sqrt{\frac{z'_1 z'_2}{z'_3 z'_4}}, \quad z_3 = \sqrt{\frac{z'_1 z'_3}{z'_2 z'_4}}, \quad z_4 = \sqrt{\frac{z'_1 z'_4}{z'_2 z'_3}}, \quad (7.17)$$

Performing the change of variables dictated by the $\mathcal{M}$ matrix resulted in a new GKZ system with the denominator corresponding to the $A_{\text{can.}}$ matrix, as expected. The parameters γ are transformed, $\gamma \rightarrow \gamma'$, also as expected:

$$\gamma' = \begin{bmatrix} \gamma'_0 \\ \gamma'_1 \\ \gamma'_2 \\ \gamma'_3 \\ \gamma'_4 \end{bmatrix} = \begin{bmatrix} \gamma_0 \\ 2\gamma_0 + \frac{\gamma_1 + \gamma_2 + \gamma_3 + \gamma_4 + 3}{2} \\ 2\gamma_0 + \frac{\gamma_1 + \gamma_2 - \gamma_3 - \gamma_4 + 3}{2} \\ 2\gamma_0 + \frac{\gamma_1 - \gamma_2 + \gamma_3 - \gamma_4 + 3}{2} \\ 2\gamma_0 + \frac{\gamma_1 - \gamma_2 - \gamma_3 + \gamma_4 + 3}{2} \end{bmatrix}. \quad (7.18)$$

8 Code

In working on this project, we developed *Feynkit*, a fully open-source Python library to simplify the calculations described throughout this text. We developed this code to remove the need for computations by hand, which become tedious even for relatively simple diagrams and integrals. Some calculations cannot even be done by hand, such as finding the toric ideals of GKZ systems, and so finding a computational solution was necessary. However, such a program was not available to us, which is why we decided to build our own.

While there already exists a codebase which can perform similar computations to ours, it is written in Mathematica, which is proprietary software [33]. Indeed, our library is the first open-source code for research into the GKZ systems of Feynman diagrams.

Our first attempt at writing code focused on one-loop integrals, using the same techniques as in Section 3. Starting from the momentum representation, it parameterised the integral into the LP representation by completing the square in the exponential. This method worked to find the GKZ integral and all the salient related information, but was greatly limited by the fact that it only applied to one-loop integrals. Built using the symbolic mathematics library Sympy [34] and mathematics software system SageMath [35], it laid the foundation for a more robust and general approach which would become Feynkit.

The final version of the software uses techniques from Lee and Pomeransky [36]. This way, starting from any Feynman diagram, no matter how complex, how many loops or edges, we can always compute its complete GKZ system, from the $\mathcal{A}$ -matrix, to its Euler equations, from the toric ideal to the shape of its polytope. The code was tested on known Feynman diagrams and gave the expected results. It is theoretically capable of finding the GKZ representation of any Feynman

diagram, however at 3 loops and greater, we could not get results as the computational time was excessive, at least on the machines available to us.

The features implemented within our library include the ability to find:

- The Feynman integral of any Feynman diagram;
- The LP representation of any Feynman integral;
- The GKZ integral for any Feynman integral;
- The toric ideal and Euler equations corresponding to any Feynman integral
- The $\mathcal{A}$ -matrix corresponding to any Feynman integral;
- The polytope corresponding to any Feynman integral, including PCA projections for lattice polytopes in higher than three dimensions of parameter space;
- The lexicographically ordered $\mathcal{A}$ -matrices;
- Volume vectors of polytopes, and therefore whether two polytopes are affinely equivalent;
- The affine transformation relating two affinely equivalent polytopes;

Furthermore, the code automatically formats all the information above into a single $\text{\LaTeX}$ file for easier comprehension.

The library can be found [here](#).

9 Discussion

9.1 The ordering problem

As we found in § 7, in the context of affine transformations, the ordering of the A -matrix of a GKZ system matters. This result runs contrary to what we would expect, as discussed in § 7.2. In two ways, the ordering of the columns of the A -matrix should not matter.

The first way is due to the definition of the A -matrix as the matrix of powers of $\mathbf{z}$ in the polynomial $\mathcal{D}$. As a reminder, each column $a_i \in A$ represents a monomial in $\mathcal{D}$. But the ordering of the polynomials of terms in $\mathcal{D}$ is arbitrary, and permuting terms results in the same polynomial, due to the commutative property of addition. Therefore, the a_i can be permuted in the same way and leave the GKZ system unchanged.

Ordering should be irrelevant to a second degree given the definition of our polytopes as a convex hull of a set of points $a_i \in A$. The convex hull is the smallest convex set that completely encloses the set. The ordering of points is not important as the convex hull stops looking at the matrix A as a whole, rather looking at its individual component columns. Each column is a point in parameter space, and swapping around columns does not change the position of that point in space. Moreover, for the reasons given above, the matrix A is permutation invariant. Therefore, according to Kim, Tawarmalani, and Richard [37], $\text{conv}\{a\}$ is an invariant of A .

So, ordering or not, both the GKZ system and the polytope are the same. Nevertheless, once we need to compute anything to do with GKZ systems and their polytopes, we must impose an order. How we choose to order the columns of A does not appear to be important, so long as it is consistent across every A that we are considering.

There does not seem to be a canonical or “natural” ordering of A , at least not one that is apparent. Ideally, we could find an ordering which could conserve some information about the GKZ system through an affine transformation, for example, the placement of monomials of the Symanzik polynomials $\mathcal{U}$ and $\mathcal{F}$ in A .

Forcing an ordering onto the GKZ system does not affect it in any way. However, for the GKZ system to be useful to us, it must be ordered. This tension implies a subtle disconnect between the integral-picture and the polytope-picture. Two resolutions are possible: first, we show that the ordering does not matter, and we can somehow compute transformations without permuting A -matrices. Conversely, we find a canonical ordering, and thus we must apply it to all A matrices. We believe that we would glean far more information from this second option, and potentially reveal a deeper connection between the GKZ integral and its polytope.

9.2 Boundary and interior points

Another subtlety, related to the search of new pairs of affinely equivalent polytopes, discussed in § 7, concerns interior and boundary points of polytopes. This is best illustrated with the example of the two-mass bubble Feynman diagram (2MB), effectively the Feynman diagrams from Figure 4 with two massive edges, m_1 and m_2 . Its corresponding LP representation polynomial is

$$\mathcal{G}_{2\text{MB}} = z_1 + z_2 + m_1^2 z_1^2 + m_2^2 z_2^2 + (m_1^2 + m_2^2 + p^2) z_1 z_2, \quad (9.1)$$

and so has as its associated A -matrix

$$A_{2\text{MB}} = \begin{bmatrix} 1 & 0 & 2 & 0 & 1 \\ 0 & 1 & 0 & 2 & 1 \end{bmatrix}. \quad (9.2)$$

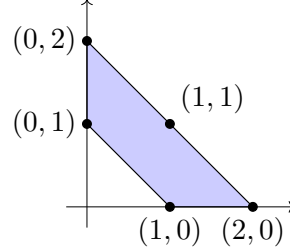


Figure 7: Polytope of the two-mass bubble Feynman diagram, $\Delta_{2\text{MB}}$, Equation 9.2.

From the polytope of the 2MB, Figure 7, we notice the point $(1, 1)$, which is directly on the boundary of the polytope on the edge connecting the points $(2, 0)$ to $(0, 2)$. As it is not a vertex of the polytope, it is omissible from the polytope, and removing it would leave the polytope's shape completely unchanged.

Consider that we do not know the GKZ representation of the 2MB. Suppose we started with a different object associated to a two-dimensional polytope $\Delta_{\text{iso. trap.}}$ which is some isosceles trapezoid (the same shape as the 2MB's polytope), described by four vertices, namely it has four columns in its A -matrix, $\{a_1, a_2, a_3, a_4\}$. We can find a transformation $\mathcal{M}$ that brings $\Delta_{\text{iso. trap.}}$ to $\Delta_{2\text{MB}}$, as these two polytopes are clearly affinely equivalent by construction.

Now, the question is whether the $\Delta_{2\text{MB}}$ that we obtain from this transformation, whose original A -matrix was composed of only four columns, is equivalent to the one with five columns, as in Equation 9.2. On the polytope side, they are the same, but on the integral side, this is not the case. Removing a column fundamentally changes the differential equations obeyed by the GKZ system, and the restriction process is now non-trivial. We could decide to add a column to the A -matrix corresponding to the missing $(1, 1)^T$ column, but this choice is only available to us since we know the answer.

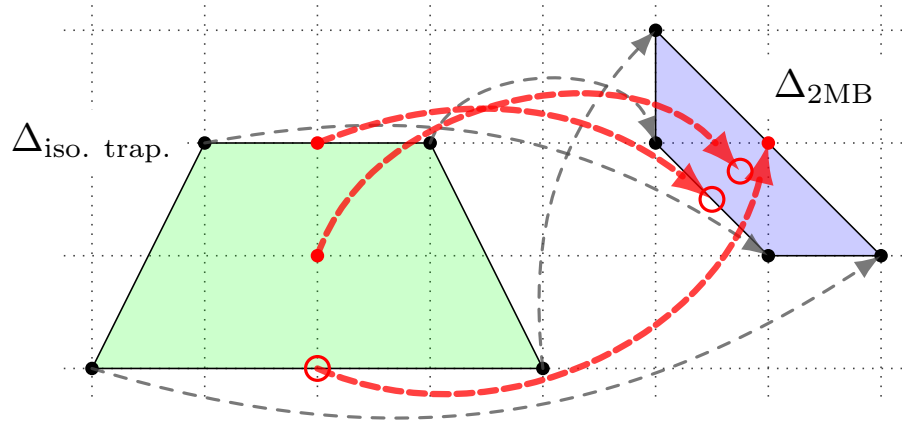


Figure 8: The transformation for some isosceles trapezium polytope, $\Delta_{\text{iso. trap.}}$, with extra points defined on it, into the 2MB polytope, $\Delta_{2\text{MB}}$. Extra boundary/interior points added and mapped in red to highlight issues with current understanding of affine transformations.

Going further, consider the polytope $\Delta_{\text{iso. trap.}}$, but this time we add an interior point to its A -matrix, by appending onto it some column vector a_5 . It is still the same polytope, due to the definition of a convex hull. Performing the same transformation as above into $\Delta_{2\text{MB}}$, we will still obtain the same shape, and the interior point will remain on the interior due to the properties of affine transformations. Even though both A -matrices now have the same dimension, whether the two

polytopes are affinely equivalent is ambiguous. This is because the extra point cannot be mapped properly onto the 2MB, as the mapped point does not exist in the 2MB's A -matrix.

In addition to all this, there is no guarantee that polytopes with these extra points on the boundary or interior even transform into a lattice polytope. These polytopes are lattice polytopes, so when transforming, every point must move to integer coordinates. This special case, and those mentioned above are shown graphically in Figure 8. Although this is a made-up example, constructed to show the contradictions at hand, it demonstrates the issues that could occur in real examples and complicate computations.

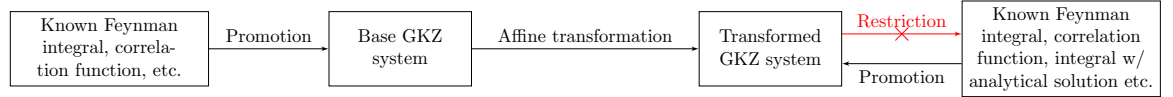


Figure 9: Flowchart showing the procedure of transforming from one known object into another one, via GKZ and affine equivalence. The final step of restriction from the transformed GKZ to the final object is not possible in general for reasons discussed in § 9, and instead the two objects must “meet in the middle”.

Now we move to think about the GKZ process on a larger scale. Imagine systematically checking for affine equivalences between different objects, say a known LP representation and integrals with a known analytical solution. The existence of boundary and interior points would break the systematic checking of affine equivalence, even if the shapes of their polytopes are indeed equivalent. Perhaps this is an issue with the volume vector method, suggesting that it must be superseded by a more robust method or analysis, but we have not encountered such a solution in the literature.

Ultimately, we have to “meet in the middle” by promoting the target object to become that transformed GKZ, rather than the other way around, restricting the GKZ to get the target. This supposes that we already know the two objects and their GKZ systems, which is, in general, not the case. The current procedure we summarise in Figure 9.

Consider the above case of two polytopes, one of a known LP representation, and the other a polytope of an integral with a known analytical solution. There are two ways that we see of resolving the issues arising from the existence of the boundary and interior points: either we remove them entirely from consideration, or accept them as part of an improved methodology.

If we accept that we can remove the extra points and consider only the vertex points, then we can say that the polytopes are affinely equivalent. If we do this, however, can we say that the GKZ systems which underlie the polytopes are also affinely equivalent? Does the affine transformation affect the underlying differential equations in the GKZ picture?

Conversely, if we determine that these polytopes are not affinely equivalent, then it follows that shape is no longer sufficient to determine affine equivalence. What extra conditions must we impose to ensure consistency of results when performing the volume vector test? What are the effects of imposing further conditions on the GKZ systems of the polytopes?

While we do not have the answers to these questions, resolving them will surely aid us to glean insight into the subtleties of the connection between the GKZ integral and its polytope.

10 Conclusion

In this report, we have introduced the concept of GKZ systems, their A -matrix and their associated polytope. We started by deriving the GKZ systems of the triangle Feynman diagram and triple K integral. From here, we derived the affine transformation that connects them, and therefore determined that they were affinely equivalent. Using this example as a guide, we showed the affine equivalence of three objects: the (deformed) massless-bubble, the one-mass bubble, and the double K integral. In doing so, we borrowed tools from metric geometry to establish affine equivalence for any two polytopes.

In the process of finding this affine equivalence, we found evidence of tension between GKZ integrals and their associated polytopes. Indeed, it seems as though there is a disconnect between the polytope picture and the GKZ picture, with reference to ordering of A -matrices and interior and boundary points.

As we have discussed above, the existence of interior and boundary points means that finding affine equivalencies between candidate polytopes becomes difficult. Our current methods, using the volume vector test which we have developed in our report, has limitations in such cases. As this test operates only on the polytope side, it is completely blind to the integral side, so questions about ordering or extra points simply cannot be answered.

Finally, underscoring all of our work presented here is the Python library that we have written. The first open-source code of its kind, it has enabled us to work with GKZ systems with considerably greater ease, eliminating laborious by-hand manipulations. This library is an important tool that will aid us and others in probing deeper into GKZ theory and a tangible outcome from this work.

The study of GKZ systems in the context of physics is still burgeoning; finding answers to these open questions will likely lead to it taking on a greater role in the progress of various fields, including particle physics, cosmology, and CFT.

10.1 Future research directions

10.1.1 Improved affine equivalence test As mentioned, our current ability to determine affine equivalence of GKZ systems is via methods that work only on the polytope side. A more comprehensive method should be developed, which takes inputs from both sides, addressing the issues of ordering and extra lattice points. If such a method were to be developed, it almost assuredly would provide us with deeper insights into the connection between the integral and polytope sides of GKZ. Furthermore, it would allow us to explore more connections between GKZ objects, with the hopes of finding more affine equivalencies between known and important mathematical objects.

10.1.2 Investigating different mathematical objects as GKZ integrals As we have highlighted throughout this report, the possibility of equating a high-order Feynman integral to some integral with a known analytical solution represents a significant opportunity. This would entirely cut out computations from perturbative calculations. Nevertheless, in order to find an affinely equivalent GKZ system to a particular system under consideration would require a larger sample of GKZ systems to compare to. As we summarised in the previous section, the only reliable method for finding affinely equivalent pairs currently available to us is “meeting in the middle”. This requires a large sample of GKZ systems to test equivalence with, something that has not yet been compiled. Constructing some sort of library of mathematical objects, with their related GKZ systems and polytopes, would be hugely advantageous for researchers in finding pairs of these affinely equivalent GKZ systems.

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